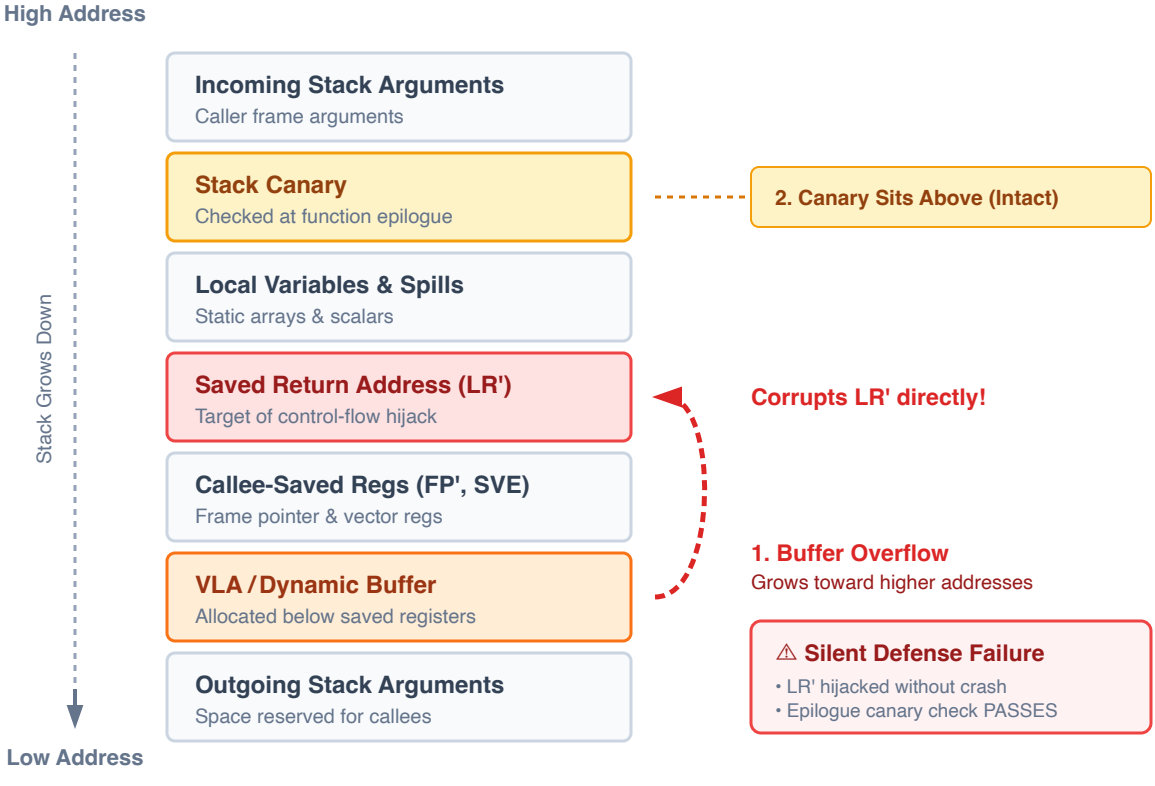

High Address
Stack Grows Down
Incoming Stack Arguments
Caller frame arguments
Stack Canary
Checked at function epilogue
Local Variables & Spills
Static arrays & scalars
Saved Return Address (LR')
Target of control-flow hijack
Callee-Saved Regs (FP', SVE)
Frame pointer & vector regs
VLA / Dynamic Buffer
Allocated below saved registers
Outgoing Stack Arguments
Space reserved for callees
Low Address
2. Canary Sits Above (Intact)
Corrupts LR' directly!
1. Buffer Overflow
Grows toward higher addresses
⚠ Silent Defense Failure
• LR' hijacked without crash
• Epilogue canary check PASSES